

Intelligent Transportation Systems (ITS) Analysis, Modeling and Simulation (AMS) Community of Practice (CoP)

Spring 2026 Meeting

April 22, 2026

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Agenda

Time (Eastern)	Topics	Speakers and Facilitators
01:00–01:10 p.m.	Welcome and Recap from Last Meeting	
01:10–01:30 p.m.	Community Member Announcements	Members
01:30–02:30 p.m.	Community Task Forces Organization, Mission, Activities (to include one 15-min breakout)	Invited Panel Speakers: • Chris Melson (Oregon DOT) • George List (NC State) • Jongsun Won (FHWA) • Ossama Ramadan (WSP)
02:30–02:40 p.m.	Break	
02:40–03:30 p.m.	Invited Speaker: Solicit Feedback for CAV Technology Tracking Resource	Claire Silverstein and Rick Grahn, Noblis
03:30–03:50 p.m.	Traffic Analysis, Modeling, and Simulation: Impacts of Connectivity and Vehicle Automation	
03:50–04:00 p.m.	Wrap Up and Adjourn	

Welcome and Recap from Last Meeting

Purpose of This Community



Bring together professionals and stakeholders engaged in ITS AMS activities.



Encourage knowledge exchange on tools, methods, and applications.



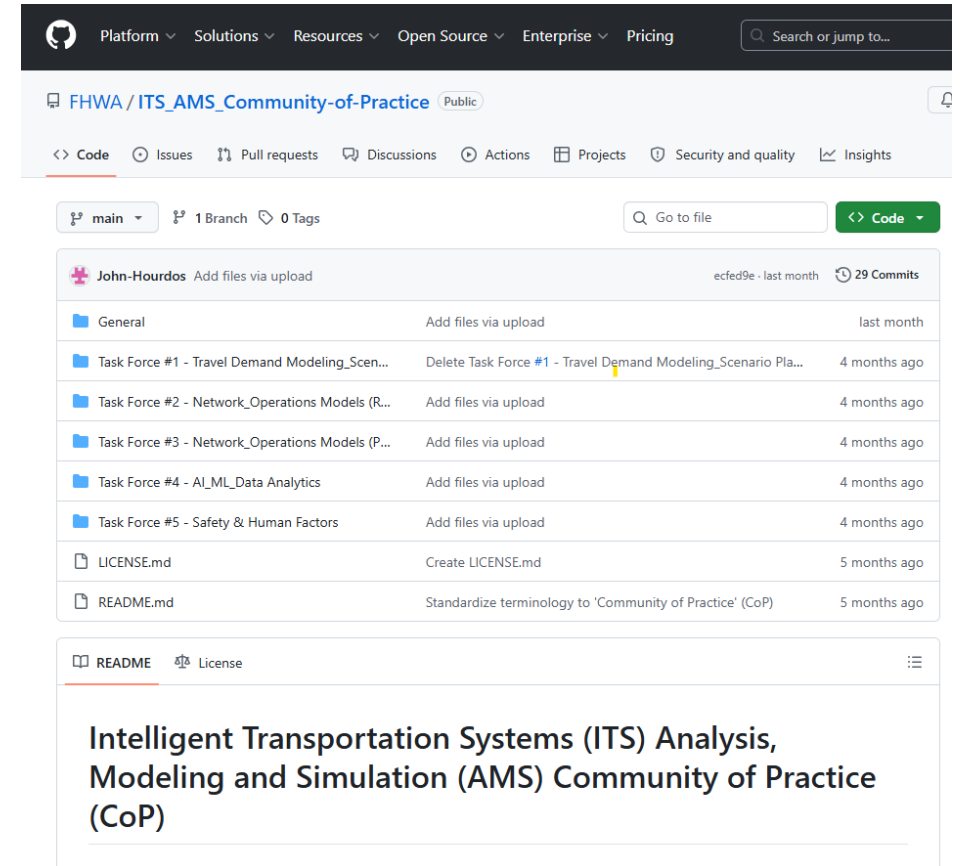
Identify shared challenges and opportunities for innovation.



Promote coordination among agencies, researchers, and practitioners.

ITS AMS CoP Forum

- Available at:
https://github.com/FHWA/ITS_AMS_Community-of-Practice
- If you have not joined the forum yet, please take a moment to add your GitHub email address to the chat so we can send you an invite in real-time.
 - If it's a personal email address that you are not comfortable putting in the public chat, please send a *private* chat containing your email address to Asean Davis (Noblis).



U.S. Department of Transportation (USDOT). 2025. "ITS Analysis Modeling and Simulation Community of Practice" (discussion group and resource library in GitHub repository). https://github.com/FHWA/ITS_AMS_Community-of-Practice, last accessed April 13, 2025.

Community Member Announcements

MEMBERS

Community Member Announcements

Opportunity for members to be given the floor for up to 3 minutes to provide updates related to:

- Research updates.
- Tools and software releases.
- Events and training opportunities.
- Industry updates.
- Other.



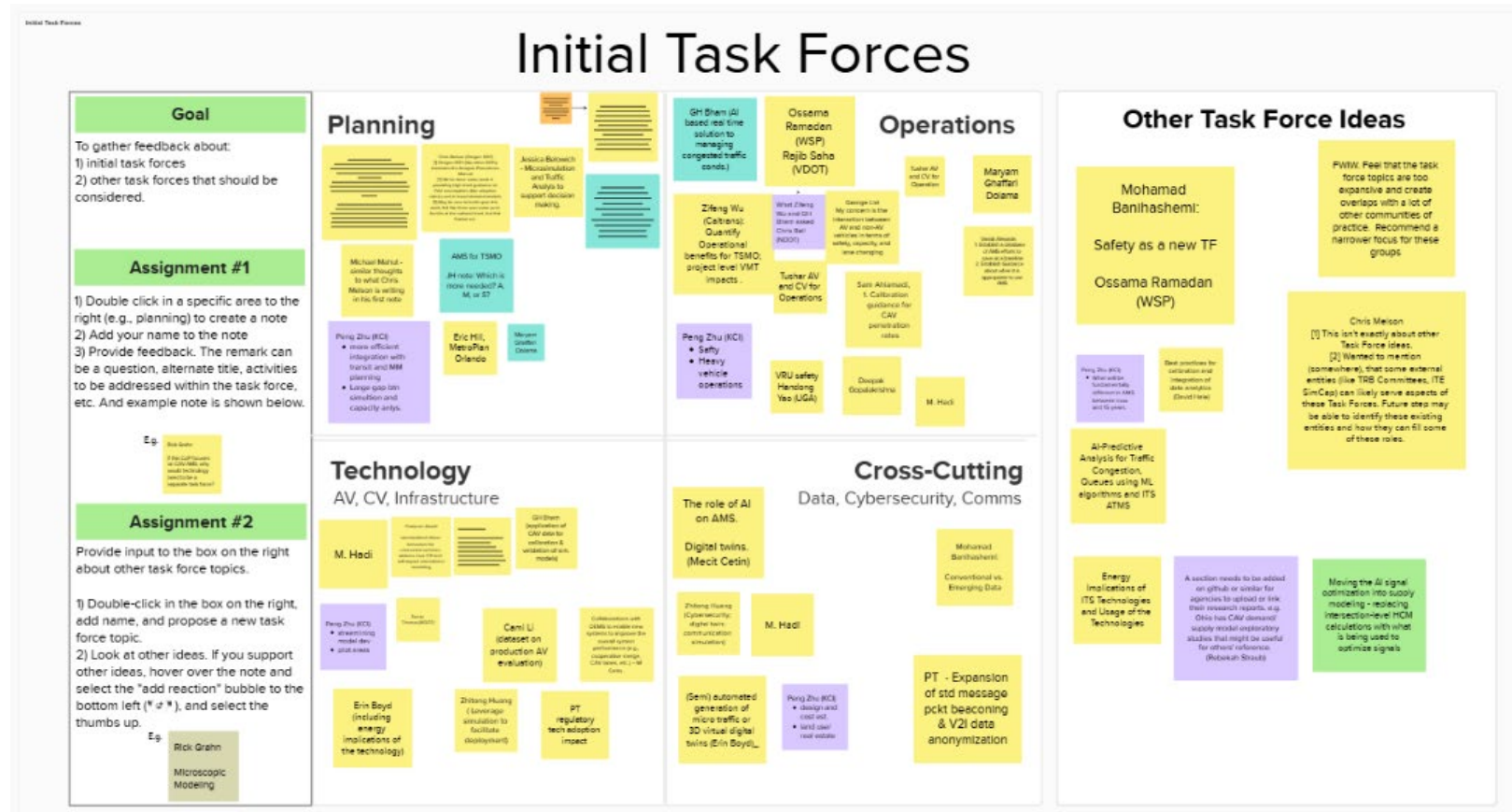
Community Task Forces Organization, Mission, Activities

INVITED PANEL

SPEAKERS:

- CHRIS MELSON, OREGON DOT
- GEORGE LIST, NC STATE
- JONGSUN WON, FHWA
- OSSAMA RAMADAN, WSP

Output from Task Force Brainstorming Activity



Selected Task Forces

- **Travel Demand /Scenario Planning:** Input assumptions (agent-based, activity based, trip-based models). Horizons, land use, penetration, pricing, fleet ownership, etc.
- **Network/Operations Models (Research):** Algorithm development for Micro-, Meso-, Macroscopic models.
- **Network/Operations Models (Practice):** Application and interpretation of Micro-, Meso-, Macroscopic models.
- **AI/ML/Data Analytics:** Traffic analysis and prediction. Optimization of fleets, operations, infrastructure.
- **Safety and Human Factors:** Human driver–machine handoffs, human driver / VRU interactions and response behaviors, etc.

Expert Insights on Task Forces



Travel Demand / Scenario Planning: Chris Melson (Oregon DOT)



Network/Operations Models (Research): George List (NC State)



Network/Operations Models (Practice): Jongsun Won (FHWA)



AI/ML/Data Analytics: John Hourdos (FHWA)



Safety and Human Factors: Ossama Ramadan (WSP)

As we hear about each task force topic, let us know which task force discussion you are interested in joining for today's breakout.



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Task Force Breakouts

Directions

- Attendees to select a breakout group to join based on a task force they are interested in to discuss:
 1. Task Force name and Mission Statement
 2. How the task force should be organized.
 3. Task force activities.
 4. How task force-specific meetings should be set up.
 5. Relationships with other groups (TRB, ITE, AASHTO, etc.)

Each group will need to select someone to report out.

Potential activities to consider for item 3:

- Webinars (presentation-format)
- Roundtables (discussion-format)
- Workshops (potentially in-person in coordination with another event like TRB)
- Peer Reviews
- Case Studies
- Tutorials
- Mentorship Program

Please enter your breakouts.

Poll: Future CoP Initiatives

- Let us know what types of activities you are interested in the CoP pursuing!

Go to
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Enter the code

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Break

Task Force Breakouts Summary

Room 1: Travel Demand /Scenario Planning

1. Task Force name and Mission Statement

- **DRAFT Mission Statement:** *Support the ITS AMS community in identifying needs related to the integration of CV, AV, and other ITS technologies into travel demand and scenario planning within ITS AMS.*
- *OK for now, will need some work in future, both title and mission statement.*

2. How the task force should be organized

need a clear lead with bandwidth, and liaisons to other task forces and external groups

3. Task force activities.

- *Collect resources first, and let the content drive what the best engagement. Shout out for roundtables, where we can get many perspectives.*

4. How task force-specific meetings should be set up. *Consider T2 grant for peer exchanges, leverage NCHRP synthesis reports*

5. Relationships with other groups (TRB, ITE, AASHTO, etc.). *Chris has some thoughts on TRB.*

Task Force Breakouts Summary

Room 2: Network/Operations Models (Research)

1. Task Force name and Mission Statement.
 - **DRAFT Mission Statement:** *Support the ITS AMS community in identifying needs related to enhanced theory and algorithm development for models to effectively integrate, calibrate, and calculate impacts of CV, AV, and other ITS technologies using existing and new AMS approaches.*
 - Okay, but update mission statement to bring in human-driven vehicle behavior and other road user behavior with mixed CAV traffic interactions, make this task force a clearinghouse to framework/clearinghouse for research – then this will be agnostic about what's out there and just reflect as an expert panel to raise the flag for reporting out gaps/issues
 - Need to focus on 1-3 topics total for now before momentum is gained
2. How the task force should be organized.
 - Ad-hoc, if we have someone to take notes, then someone to organize discussion, enough structure
3. Task force activities.
 - e.g., Webinars (presentation-format), Roundtables (discussion-format), Workshops (potentially in-person in coordination with another event like TRB), Peer Reviews, Case Studies, Tutorials, Mentorship Program, etc.
 - All of these are good, but could look to fill identified gaps, could target group of states interested in sponsoring a pooled fund study in the future
4. How task force-specific meetings should be set up. – in conjunction with the broader CoP meetings
5. Relationships with other groups (TRB, ITE, AASHTO, etc.). – more TRB and ITE, maybe NCHRP, less AASHTO

Task Force Breakouts Summary

Room 3: Network/Operations Models (Practice)

1. Task Force name and Mission Statement.
 - a. **DRAFT Mission Statement:** *Support the ITS AMS community in determining the correct application and interpretation of models (micro, macro, meso) to effectively integrate, calibrate, and calculate impacts of CV, AV, and other ITS technologies using existing and new AMS approaches.*
 - b. Change “correct” to “appropriate”.
 - c. Focus: How to get this information/models to the broader practitioner group. And how to use existing tools that practitioners use today.
 - d. Focus: Also need to recognize the limitations of the tools that we have today– and communicate to tool users. Need to complement guidance about inputs/assumptions with limitations/uncertainty.
2. How the task force should be organized.
3. Task force activities.
 - a. e.g., Webinars (presentation-format), Roundtables (discussion-format), Workshops (potentially in-person in coordination with another event like TRB), Peer Reviews, Case Studies, Tutorials, Mentorship Program, etc.
 - b. Industry outreach – OEMs/ADS developers to better understand the technology capabilities (e.g., latency, feedback time)/behaviors. How to more realistically model these technologies → then communicate info to practitioners.
 - c. Peer review would be interesting
4. Relationships with other groups (TRB, ITE, AASHTO, etc.). *ITE simulation and capacity analysis committee.*
 - TRB
 - o B0194 - Committee on Intersection and Roadway Capacity and Quality of Service (fka HCQS)
 - o ACF11 - Standing Technical Committee on Traffic Flow Theory, Models, and Simulation
 - ITE
 - o Simulation and Capacity Analysis Committee (SimCap)
 - AASHTO
 - o Committee on Transportation System Operations
 - o Committee on Traffic Engineering
 - o Pooled Fund Study – Traffic Analysis Roundtable

Task Force Breakouts Summary

Room 4: AI/ML/Data Analytics

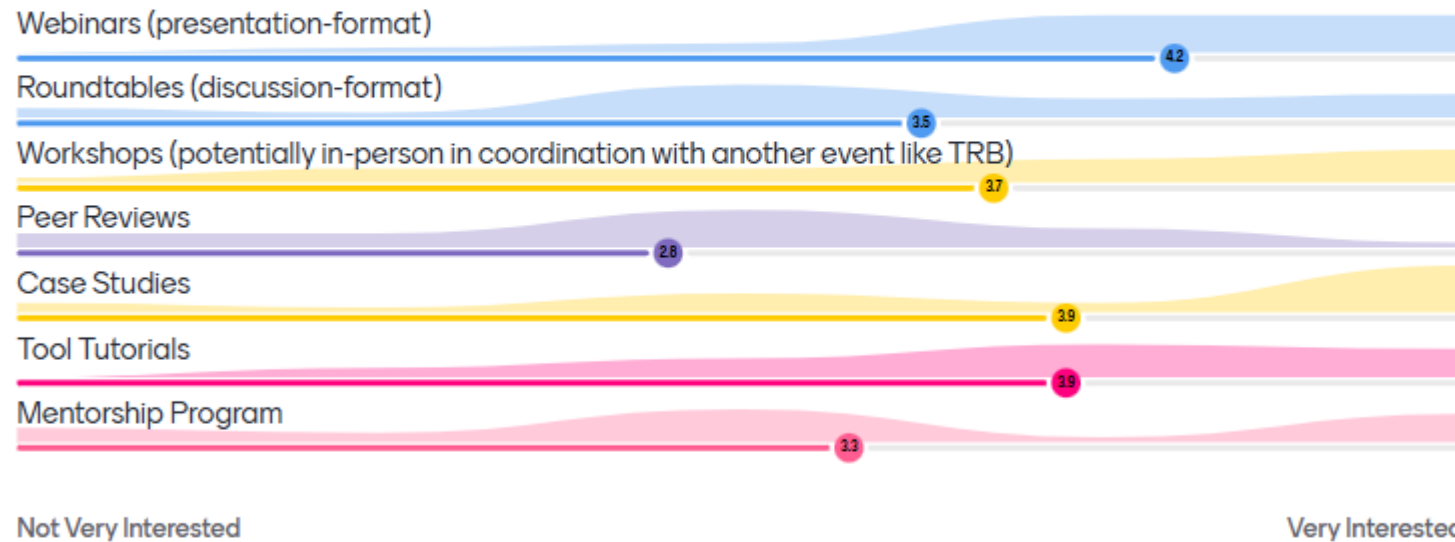
1. Task Force name and Mission Statement.
 - **DRAFT Mission Statement:** *Support the ITS AMS community in exploring how AI/ML can strengthen the phases of the modeling process for CV, AV and other ITS technologies, helping address current limitations and manage the growing complexity of modeling systems.*
 - Expand the mission statement to include all AMS approaches including conventional simulation approaches and human driven vehicle considerations.
2. How the task force should be organized.
3. Task force activities.
 - e.g., Webinars (presentation-format), Roundtables (discussion-format), Workshops (potentially in-person in coordination with another event like TRB), Peer Reviews, Case Studies, Tutorials, Mentorship Program, etc.
 - Once we determine the final activities, we could move backwards to determine the best organization
4. How task force-specific meetings should be set up.
 - Like the current setup with open discussion
5. Relationships with other groups (TRB, ITE, AASHTO, etc.).
 - Connection with ITE, ASASHTO, SITSO? , Goal to share with SIMCUP. If there are specific needs for outreach with other groups the team can work on creating material.

Task Force Breakouts Summary

Room 5: Safety and Human Factors

- **Task Force name and Mission Statement**
 - Since human factors is not a primary focus of safety AMS, consider having the title just be “Safety AMS”. Clarify that this is surface transportation, not open to other modes.
- **How the task force should be organized.**
 - Support having a chair or co-chair. Want to have separate quarterly meetings (can move to monthly cadence only if preparing for an upcoming event).
- **Task force activities.**
 - First initiative will be recruiting more members. Interested in potentially doing webinars or workshops.
- **Relationships with other groups (TRB, ITE, AASHTO, etc.)**
 - Connections through the following:
 - ITE Data Driven Safety Analysis Committee (Ossama is chair)
 - TRB Transportation Safety Management and Performance Committee
 - Human Factors Ergonomics Society (HFES)
 - TRB Human Factors Committee

What kind of activities would you like to see this CoP



Wrap Up and Adjourn

JOHN HOURDOS, FHWA

Next Community Meeting

- Anticipated for Summer 2026.
- Would any members be interested in being a featured speaker? If yes, let us know and we will coordinate with you.



Contact

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Thank You!